1.2 INTRODUCTION OF LT/ HT LINE

Generated power from power station is transmitted and is distributed through transformers, overhead lines and cables to the end users.

1.2.1 LT Lines

In India, low Tension (LT) supply is 440 volts for three-phase connections and 230 volts for single-phase connection. Consumer of electricity like individual houses, shops, small offices and smaller manufacturing units get their electricity on LT connection.

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Fig 1.2 *LT Line*

1.2.2 HT Line voltage

High tension (HT) supply is applicable for bulk power purchase which needs 11 Kv or above. Major industries are operating at High Tension supply only.



Fig 1.3 *HT Lines*

1.3 ELECTRICAL SAFETY AND PRECAUTIONS

A man who works in the electrical appliances must handle the work carefully without any damage to the equipment and workers. They must know all the operations of electrical equipment clearly. Electrical accident may occur only due to carelessness. Due to this, workers will get injured and cause damages. To avoid this, electrical workers must follow the rules and regulations while working.

1.3.1 Steps to be followed while handling Electrical Equipments

1. Before the use of equipment, one must know about the complete operation of the equipment. Electrical connections are made properly according to the connection diagram.
2. Only the skilled person is allowed to operate, testing and repairing machines.
3. A person who works in the electric post and tower post must wear safety belt and gloves.
4. After earthing, the overhead lines can be discharged by discharge rod.
5. Check the condition of all hand tools.
6. While changing the fuse wire, keep the main switch 'OFF'. Use proper rating of fuse wire for replacement.

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